UNIVERSITY OF CAPE COAST
DEPARTMENT OF MATHEMATICS
MAT 101: ALGEBRA AND TRIGONOMETRY

TUTORIAL SET 4.

1.(a) Given that $log_b a + log_c a = 2log_b a log_c a$, express a in terms of b and c.

(b) Solve for x if $log(x+3) + logx = 1$.

2. Let $log_2 x = k$, show that $log_{2x} 2 = \frac{1}{1+k}$.
Hence or otherwise, solve the equation $log_{2x} 8 - 3log_2 2x + 6 = 0$.

3.(a) Expand $[1 + (\frac{1}{2}x + x^2)]^4$ by ascending powers of x up the term in x^5.

(b) Evaluate $(1.0104)^4$ by putting $x = 0.02$ in the expansion in (a), correct to five decimal places.

4.(a) Decompose the function $f(x) = \frac{2x}{x^2-4}$ into partial fraction.

(b) Hence, find the expansion of $f(x)$ as far as the term in x^5.

(c) Show that $f(x)$ may be define as $f(x) = \frac{-1}{2} \sum_{r=0}^{\infty} \frac{x^{2r+1}}{4^r}$.

5.(a) Decompose the function $g(x) = \frac{3+x}{(1+x)^2}$ into partial fraction.

(b) Expand $\frac{1}{(1+x)}$ and $\frac{2}{(1+x)^2}$.

6.(a) Use the binomial theorem to expand $(\frac{a}{3} - 2b^3)^5$

(b) Show that $\frac{x^n + x^{n+2}}{x+x^3} = x^{n-1}$.